Retrieving People and Objects from Textual Descriptions via Multimodal Learning

HK252-DATN-160 - Capstone Project

Committee 6A Computer Science
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Introduction

Motivation: Relation-Aware Understanding

Text-based image retrieval becomes difficult when the query describes **relations**, not only object categories.

Application value

- Relation triplets such as (person, riding, horse) support retrieval/localization from natural-language descriptions.
- Applications include security search, robot commands, AR guidance, and e-commerce search.

Technical challenges

- Directional relation semantics, dense visual occlusions, fine-grained grounding, and hallucinations.
- Long-tailed predicate imbalance, hidden rare-relation failures, and out-of-distribution.

Community context

- Recent CVPR/ICCV work keeps pushing SGG toward debiasing, query-based decoding, and open-vocabulary prediction.
- Standard datasets such as Visual Genome make protocol-compatible comparison possible.

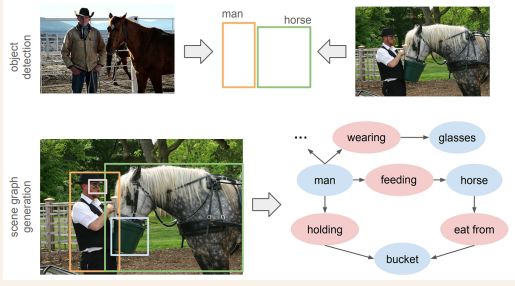
Source: Application and protocol context: Visual Genome, Neural Motifs, TDE.

Scene Graph Generation

Task formulation

Input: image I, object boxes $B = \{b_i\}_{i=1}^{N}$, and labels $O = \{o_i\}_{i=1}^{N}$ in PredCls.
Output: ranked directed triplets

$$\mathcal{G}(I) = \{(s_i, p_{ij}, o_j, \sigma_{ij}) \mid i \neq j\}.$$



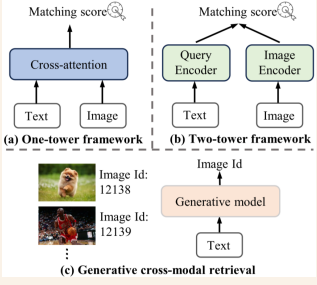
Source: SGG overview image and VG protocol context: Visual Genome and Neural Motifs.

Relation-Aware Retrieval

Triplet-index retrieval

Input: a text query q, e.g., “man ride horse”.
Output: images ranked by triplet-query agreement:

$$\text{score}(I, q) = \max_{(s,p,o)\in\mathcal{G}(I)} \text{sim}\big(q, (s, p, o)\big).$$



Source: Retrieval formulation based on predicted scene-graph triplets; structured SGG motivation follows SGTR.

Objectives and Scope

Objectives

1. Build **PURE**: a compact directed-pair predicate model using dense visual evidence.
2. Evaluate predicate ranking with both aggregate recall and mean recall: R@K and mR@K.
3. Build **CORE**: a controlled diagnostic benchmark for relation sensitivity and retrieval stress tests.
4. Demonstrate retrieval as a direct application of predicted relation triplets.

In-scope work

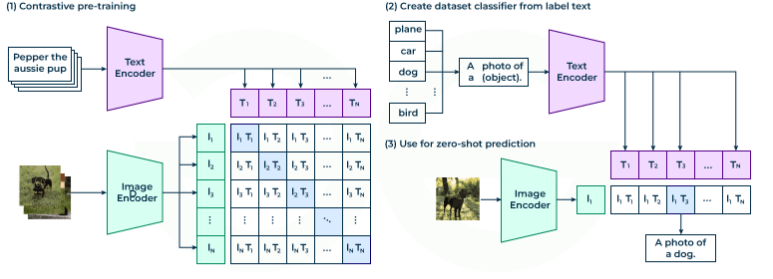
1. Vision-language relation scoring under the PredCls setting.
2. Standardized Visual Genome dataset and custom CORE benchmark.
3. Image retrieval demonstration via predicted relation triplets.
4. Recent SGG architectures used as methodological context, not full reimplementation targets.
5. Open-source components for computational feasibility and reproducibility.

Source: Scope follows the standard separation between PredCls, SGCls, and SGDet in Neural Motifs.

Related Works

CLIP: Vision-Language Pretraining

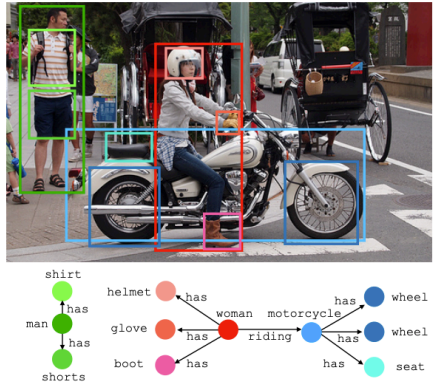
CLIP aligns image and text representations; PURE uses this shared space but adds directed relation states and predicate-specific scoring.



Source: Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021.

Neural Motifs: Context and Frequency Bias

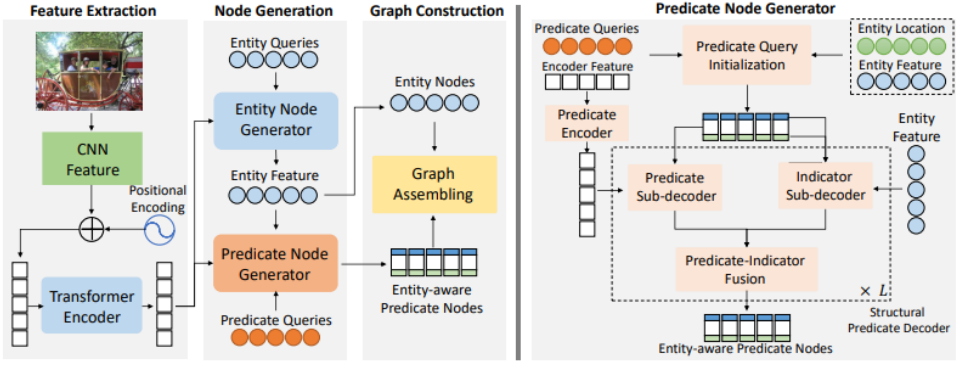
Neural Motifs defines a central VG-style SGG reference; it also shows why context and frequency priors must be separated from raw predicate scoring.



Source: Zellers et al., “Neural Motifs: Scene Graph Parsing with Global Context,” CVPR 2018.

SGTR: Transformer-Based Relation Queries

SGTR treats SGG as structured set prediction with relation queries; PURE keeps iterative relation refinement in a compact PredCls / GT-pair scorer.



Source: Li et al., "SGTR: End-to-End Scene Graph Generation With Transformer," CVPR 2022.

Other Works I: Debiasing Long-Tail Predicates

Why SGG needs debiasing

Visual Genome predicates are highly imbalanced: common relations such as “on”, “has”, or “wearing” dominate training, while rare but informative predicates receive fewer examples.

- **TDE** separates visual evidence from dataset frequency bias at inference time.
- **CFA** creates compositional feature variants to strengthen rare predicate learning.
- **SBG** estimates sample-level bias, so hard or unusual samples receive different correction.

Implication for PURE

PURE reports both raw and calibrated scores: raw scores show learned visual relation evidence, while calibrated scores show the deployable ranking after frequency correction.

Source: Sources: TDE, CFA, SBG.

Other Works II: Query-Based & Open-Vocab SGG

Structured relation prediction

Transformer-based SGG models such as SGTR and RelTR move away from classifying independent object pairs only. They use relation queries to predict structured subject–predicate–object outputs.

- This improves end-to-end graph generation, but makes training sensitive to assignment and sparse relation annotations.
- Recent Hydra-style SGG methods address this by giving one ground-truth relation multiple positive query matches.

Open-vocabulary direction

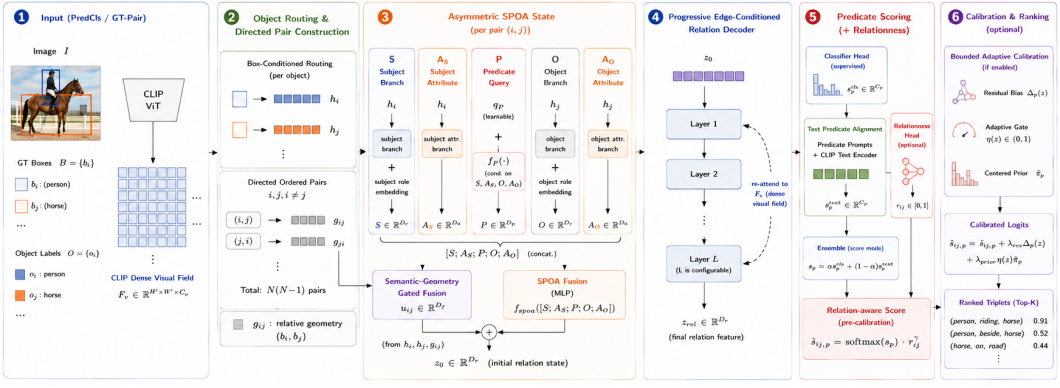
VLM-based SGG uses CLIP/LLM text spaces to score predicates beyond a fixed VG150 vocabulary. This motivates PURE's text-alignment head and the CORE diagnostic benchmark.

Source: Sources: SGTR, RelTR, Hydra-SGG, and open-vocabulary SGG/VLM literature.

Proposed Method

PURE Overview

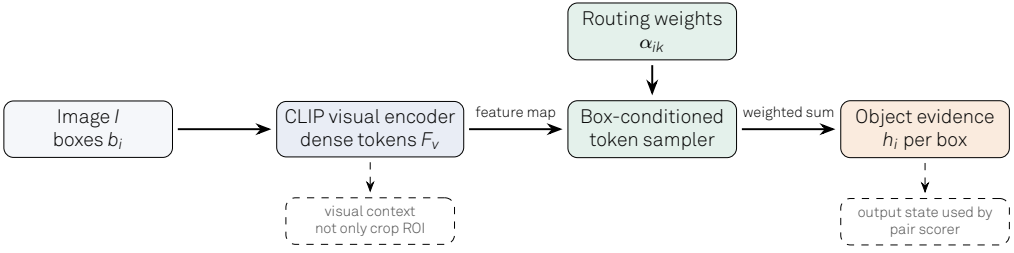
PURE is a **relation scorer**: given object boxes and labels, it ranks possible predicates for every directed object pair.



Source: PURE architecture from project implementation and assets; CLIP backbone context follows CLIP.

From Image to Object Evidence

For each detected/ground-truth box, PURE gathers visual evidence from dense CLIP features instead of relying only on a cropped ROI.



Object-routing equation

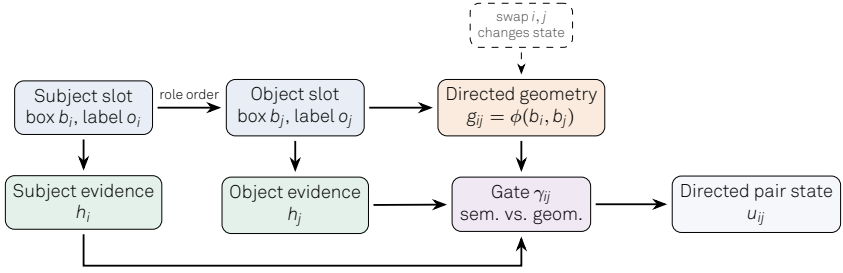
$$h_i = \sum_{k=1}^{K} \alpha_{ik} F_v(p_{ik}), \qquad \sum_{k=1}^{K} \alpha_{ik} = 1.$$

h_i is box-level evidence; $F_v(p_{ik})$ is a dense CLIP token; α_{ik} is its learned importance.

Source: Project architecture diagram and implementation component: CLIP-based `DeformableObjectRouter`.

Directed Pair Fusion

Direction matters: “person riding horse” is not “horse riding person”, so PURE builds a directed state for each ordered pair.



Semantic–geometry fusion

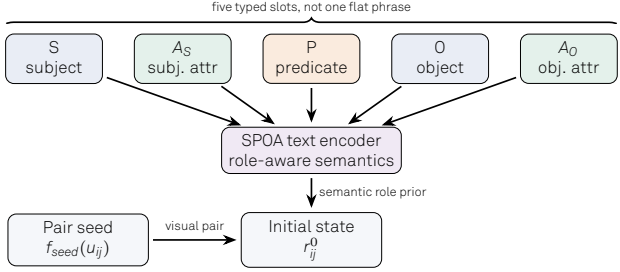
$$g_{ij} = \phi(b_i, b_j), \qquad u_{ij} = \gamma_{ij}\,\mathrm{norm}(h_i + h_j) + (1 - \gamma_{ij})\,\mathrm{norm}(g_{ij}).$$

g_{ij} is directed geometry; u_{ij} is pair evidence; γ_{ij} balances semantics and layout.

Source: Project architecture diagram and implementation components; directed relation modeling follows Visual Genome.

PURE: Asymmetric SPOA State

SPOA stores relation evidence as **Subject**, subject **Attribute**, **Predicate**, **Object**, and object **Attribute**. Attributes help distinguish entities such as “man in red shirt” versus “man in black jacket”.



SPOA initialization

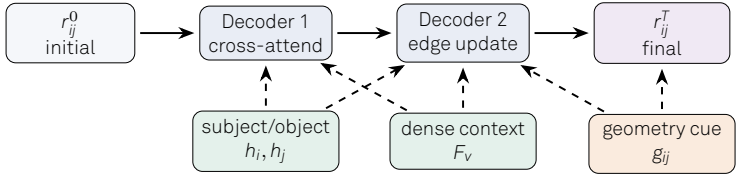
$$r_{ij}^0 = f_{seed}(u_{ij}) + f_{spoa}([S; A_S; P; O; A_O]).$$

r_{ij}^0 is the initial state. The first term uses directed pair evidence; the second term injects role-aware SPOA semantics.

Source: SPOA state from current PURE implementation; relation directionality follows Visual Genome.

PURE: Progressive Decoder

The initial relation state is refined for several steps. At each step, it can re-attend to subject evidence, object evidence, dense visual context, and geometry.



Refinement step

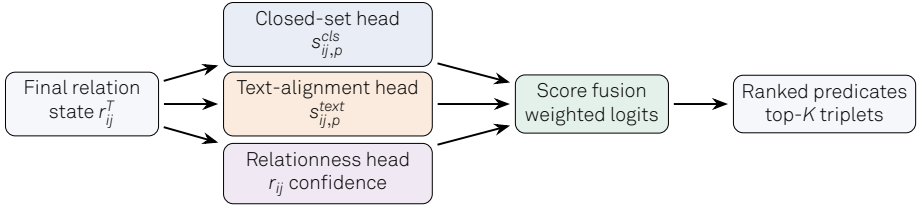
$$r_{ij}^{t+1} = \mathrm{LN}\left(r_{ij}^t + \Delta_t(r_{ij}^t, h_i, h_j, g_{ij}, F_v)\right).$$

The update Δ_t is a small correction from subject/object features, geometry, and dense visual context; layer normalization keeps the state stable.

Source: Project architecture diagram and implementation component: `ProgressiveEdgeConditionedLayer`; iterative SGG context follows IMP.

PURE: Predicate Scoring Heads

PURE does not rely on one head only. It combines a closed-set classifier, a text-alignment score, and a relationness confidence.



Predicate score

$$s_{ij,p} = \alpha s_{ij,p}^{cls} + (1-\alpha) s_{ij,p}^{text} + \beta \log r_{ij}.$$

The classifier scores VG150 labels; the text head compares against predicate prompts; relationness r_{ij} down-weights invalid object pairs.

Source: Project inference diagram and implementation components; text alignment context follows CLIP.

Training Objective

Total loss

$$\mathcal{L}_{PURE} = \lambda_{ce}\mathcal{L}_{PredCE\text{-}LA} + \lambda_{spoo}\mathcal{L}_{CF\text{-}SPOA} + \lambda_{g}\mathcal{L}_{Ground}.$$

- $\mathcal{L}_{PredCE\text{-}LA}$: predicate cross-entropy with logit adjustment for VG150 imbalance.
- $\mathcal{L}_{CF\text{-}SPOA}$: aligns visual relation states with correct text triplets and separates counterfactual negatives.
- $\mathcal{L}_{Ground}$: keeps relation evidence connected to dense visual tokens.

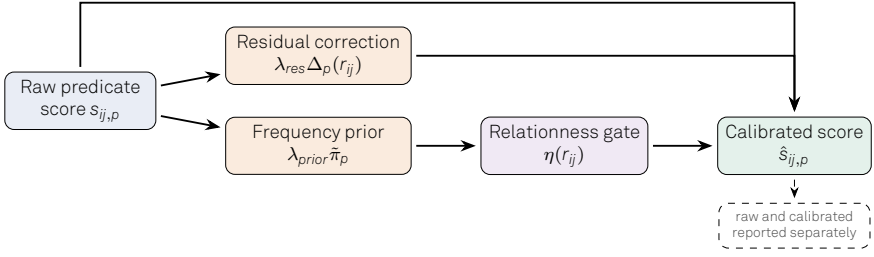
Reading

Classification gives accuracy, SPOA gives role sensitivity, and grounding reduces visually unsupported head-predicate shortcuts.

Source: Losses from current PURE implementation; long-tail motivation follows TDE.

Calibration: Evaluation-Time Score Correction

Calibration adjusts final predicate scores after training. It can boost or suppress predicates using learned residuals and frequency-prior information, but it does not change the trained model weights.



Calibrated score

$$\hat{s}_{ij,p} = s_{ij,p} + \lambda_{res}\Delta_p(r_{ij}) + \lambda_{prior}\eta(r_{ij})\tilde{\pi}_p.$$

$s_{ij,p}$ is the raw score; residual and prior terms correct ranking bias at evaluation time. Raw and calibrated results are therefore reported separately.

Source: Project calibration plot and current PURE sweeps; long-tail calibration context follows TDE.

CORE: Diagnostic Purpose

CORE focuses on relation-sensitive retrieval cases generated from controlled counterfactual prompts.

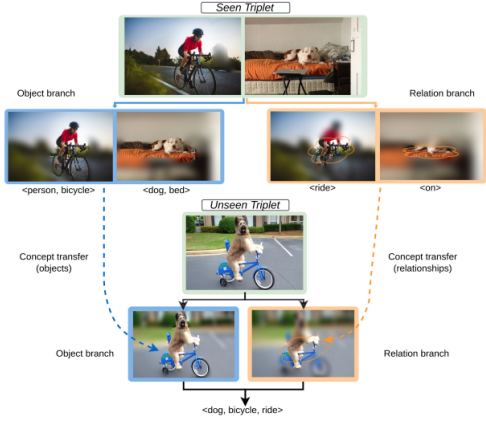
What CORE tests

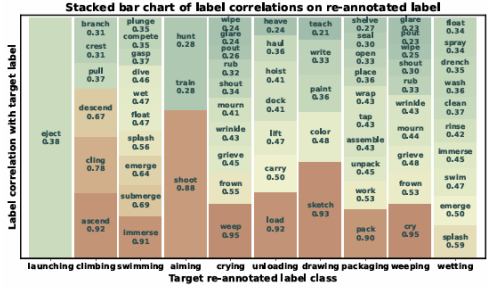
- Role sensitivity: whether subject and object are interpreted correctly.
- Attribute and spatial binding: whether modifiers and geometry attach to the right entity.
- Coverage: whether VG-compatible predicates are sufficient for diagnostic relations.

Source: CORE is project-built; compositional evaluation motivation follows SugarCrepe.

CORE Overview

CORE stands for Counterfactual Multi-label Relation Evaluation for Open-Vocabulary in the report assets; it is presented here as a controlled relation diagnostic resource.

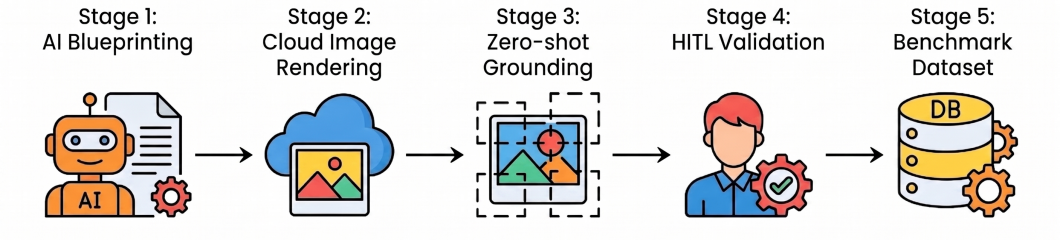




Source: CORE overview and vocabulary-similarity distribution from project analysis; compositional evaluation motivation follows SugarCrepe.

CORE Generation Pipeline

The generation pipeline combines prompt design, synthetic images, box extraction, and human/VLM audit before evaluation.



Source: Project CORE generation pipeline; open-set grounding context follows Grounding DINO.

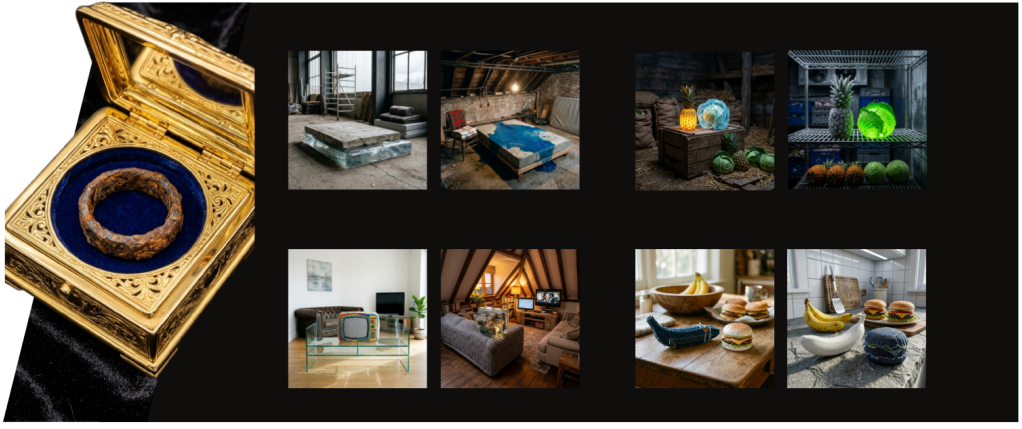
CORE Example: Action Role Reversal

This group probes who performs the action and who receives it.

Source: CORE diagnostic group from project benchmark; compositional evaluation context follows SugarCrepe.

CORE Example: Attribute Binding

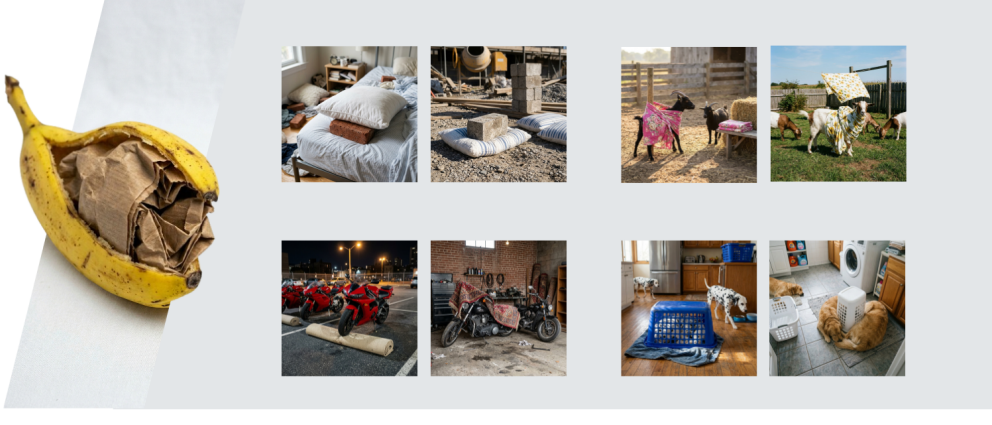
Attribute cases test whether modifiers attach to the intended entity.



Source: CORE diagnostic group from project benchmark; compositional evaluation context follows SugarCrepe.

CORE Example: Spatial and Containment

Spatial cases test support, containment, and relative layout beyond object co-occurrence.



Source: CORE diagnostic group from project benchmark; compositional evaluation context follows SugarCrepe.

CORE Example: Gaze and Attention

Gaze cases test whether attention direction is grounded in the subject-object pair.

Source: CORE diagnostic group from project benchmark; compositional evaluation context follows SugarCrepe.

CORE Example: Occlusion and Depth

Depth and occlusion cases reveal whether spatial evidence is used directionally.

Source: CORE diagnostic group from project benchmark; compositional evaluation context follows SugarCrepe.

CORE Example: Extreme Compositional OOD

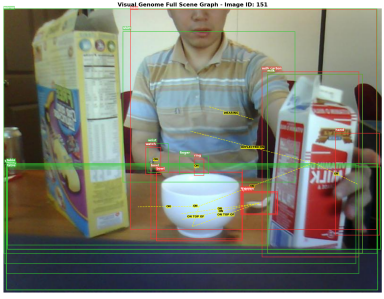
Extreme OOD cases test unusual object-relation compositions with limited VG-compatible coverage.

Source: CORE diagnostic group from project benchmark; compositional evaluation context follows SugarCrepe.

Experiments and Discussion

VG150 Metadata

VG150 is the main evaluation dataset. It is derived from Visual Genome and uses a fixed SGG vocabulary for protocol-compatible PredCls evaluation.



Metric	Value
Total images	108,077
Total regions	5,408,689
Avg. regions per image	50.04 ($\sigma = 9.25$)
Avg. coverage	548.61%
Vocabulary size	94,244
Avg. description length	5.0 words
Word entropy	5.8953
Bigram entropy	9.4997
Sampled IoU	0.0593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  ]
}
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Source: Dataset: Visual Genome; VG150 protocol: Neural Motifs.

VG150 Metadata

Evaluation metadata

- Input annotations: object boxes B, object labels O, and relation triplets (s_i, p_{ij}, o_j).
- Main protocol: PredCls / GT-pair, isolating predicate ranking from object detection.
- Main concern: predicate frequencies are long-tailed, so R@K and mR@K must be read together.

Item	Setting
Dataset source	Visual Genome, standardized as VG150 for SGG.
Input annotations	Ground-truth boxes B, object labels O, and relation triplets (s_i, p_{ij}, o_j).
Main protocol	PredCls / GT-pair: evaluate predicate ranking with objects fixed.
Main metrics	R@50 for aggregate recall and mR@50 for long-tail predicate balance.
Main readout	Raw classifier scores and calibrated scores are reported as separate operating points.

Source: Report experimental setup; VG150 protocol follows Neural Motifs.

VG150 and Metrics

VG150 is derived from Visual Genome and uses a restricted set of frequent object and predicate classes for SGG evaluation. Because predicates are long-tailed, one aggregate metric is not enough.

Recall at K

$$R@K = \frac{\#\{\text{ground-truth triplets recovered in top } K\}}{\#\{\text{ground-truth triplets}\}}.$$

R@K measures aggregate ranking quality but can be dominated by frequent predicates.

Mean Recall at K

$$mR@K = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} R_p@K.$$

Mean recall is the main long-tail diagnostic because each predicate contributes equally.

Source: Visual Genome dataset: Krishna et al., IJCV 2017; protocol follows Neural Motifs.

Training Configuration

The maintained run uses the current L4 script with dense CLIP features and relation-specific heads enabled.

Configuration	Setting
Hardware preset	NVIDIA L4 24GB, single GPU.
Precision	AMP bf16, channels-last layout, TF32 when available.
Optimization	6 epochs, batch 12, accumulation 2, gradient checkpointing.
Routing budget	Input 448, max objects 32, max pairs 64, learned prune $K = 64$.
Evaluation	PredCls / GT-pair with raw and calibrated score readouts.

Source: Maintained script: `scripts/train_l4_phase34.sh`; CLIP adaptation context follows CLIP.

Training Schedule and Presets

Preset / step	Where used	Meaning
Warmup	Training	Keep CLIP stable while relation heads and SPOA alignment start learning.
Adapt-light	Training + calibration	Controlled adaptation for relation-sensitive features and fixed-prior sweeps.
Ultra-light	Raw baseline	Low-VRAM preset to test relation-head behavior with minimal adaptation.
Prior-only	Ablation	Measures how much improvement comes from statistical prior scoring.
Calibration sweep	Evaluation	Fixed-prior calibration strengths applied to logits at inference.

Source: Report training-schedule and run-preset tables; calibration schedule context follows TDE.

Main PredCls Results

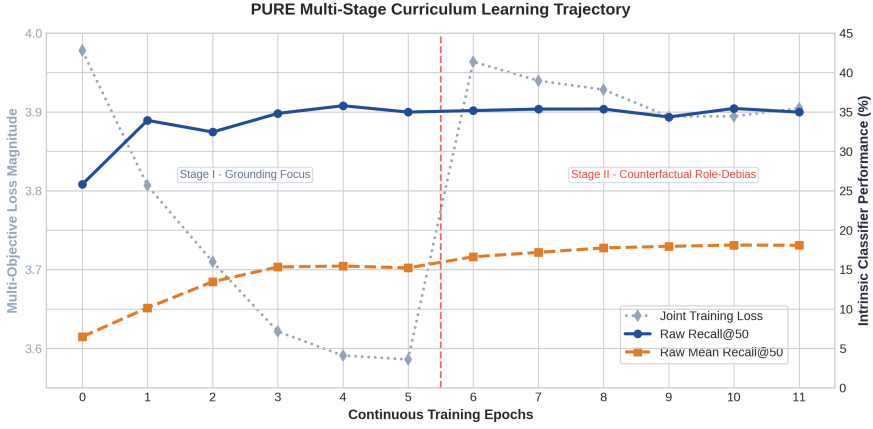
Calibration strongly improves aggregate R@50. Raw runs remain important because they reveal what the learned predicate scorer can do before score redistribution.

Configuration	R@50	mR@50	Reading
Adapt-light + fa45	**67.20**	22.74	Best aggregate recall.
Adapt-light + fa375	67.09	**22.99**	Best calibrated mean recall.
Ultra-light raw e2	35.22	20.35	Strong raw tail behavior.
Adapt-light raw e5	33.71	20.03	Tail improves with later epoch.
Prior-only raw e5	33.50	19.50	Prior baseline below Adapt-light.

Source: Project VG150 PredCls logs; dataset/protocol: Visual Genome, Neural Motifs.

Training Loss Plot

Training curves support the later ablation reading: predicate classification, SPOA alignment, and grounding stabilize together during optimization.



Source: Project training-loss plot; optimization context follows CLIP fine-tuning practice.

Calibration Strength Sweep

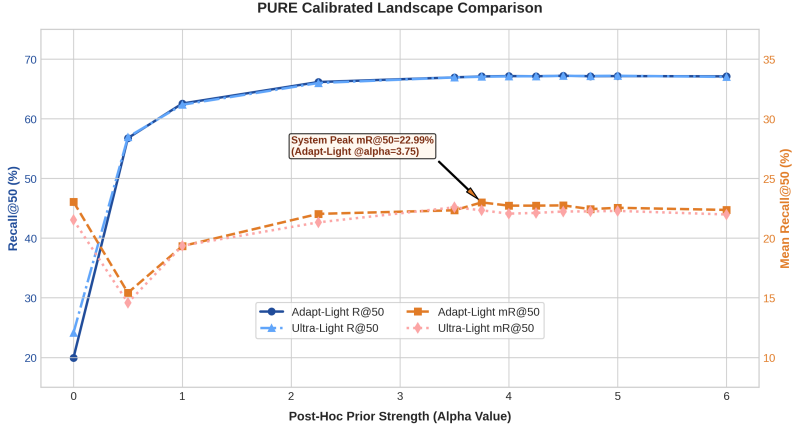
The sweep shows why calibration must be reported explicitly: the best raw tail score and best calibrated system score are different operating points.

Calibration setting	R@50	mR@50	Reading
Adapt-light raw	19.95	**23.04**	High tail evidence, low aggregate.
Adapt-light fa10	62.55	19.33	R@50 begins to recover.
Adapt-light fa225	66.15	22.03	Balanced calibrated point.
Adapt-light fa375	67.09	**22.99**	Best calibrated mR@50.
Adapt-light fa45	**67.20**	22.74	Best R@50.

Source: Project calibration sweep; long-tail calibration context: TDE.

Calibration Plot

The calibration plot visualizes the raw-versus-calibrated trade-off: aggregate recall and mean recall peak at different operating points.



Source: Project calibration plot; long-tail calibration context follows TDE.

Comparison with Related SGG Methods

PURE is strong in aggregate recall, while specialized debiasing methods remain stronger on long-tail mean recall.

Method	R@50	mR@50	Interpretation
Motifs + CFA	54.10	**35.70**	Best long-tail reference.
SBG	55.80	33.30	Strong sample-level debiasing.
ResCAGCN + PUM	N/A	20.20	Prior debiasing reference.
PURE fa45	**67.20**	22.74	Highest aggregate recall.
PURE fa375	67.09	22.99	Highest PURE mean recall.

Source: External baselines: CFA, SBG, TDE/PUM. PURE: project logs.

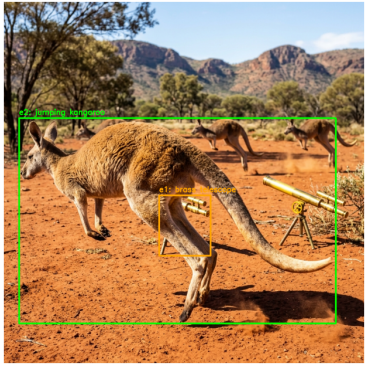
Loss Component Ablation

Removing grounding increases aggregate recall but lowers mR@50, suggesting that dense grounding helps protect tail predicates from head-prior domination.

Setting	R@50	mR@50	ΔmR	Reading
Full objective	21.17	**22.20**	N/A	Best head-tail balance.
Without SPOA	21.39	21.41	-0.79	Mild tail degradation.
Without grounding	**28.36**	19.11	-3.09	Head-prior spike; tail collapse.

Source: Project loss-ablation logs; SGG protocol follows Neural Motifs.

CORE Metadata

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    "layout_priors": "Kangaroo in foreground. Telescope in background."
  },
  "file_path": "./image/Occlusion_Depth_350_B.jpeg",
  "grounding": [
    {
      "entity_id": "e2",
      "box": [
        [0.48213905096054077, 0.6058117747306824, 0.878934383392334, 0.567793071269989]
      ],
      "visibility_status": "full",
      "multi_label_phrases": ["jumping kangaroo", "the animal in front", "kangaroo"]
    },
    {
      "entity_id": "e1",
      "box": [
        [0.5021708011627197, 0.6191258430480957, 0.14215587079524994, 0.16719689965248108]
      ],
      "visibility_status": "partial",
      "multi_label_phrases": ["brass telescope", "hidden telescope"]
    }
  ],
  "audit_metrics": {
    "vlm_model_used": "gemini-2.5-pro",
    "match_own_caption_score": 0.94,
    "match_cross_caption_score": 0.06,
    "both_entities_visible": true,
    "audit_status": "SUCCESS"
  }
}
```

CORE Retrieval Diagnostics

CORE retrieval diagnostics should be read together with coverage because each group stresses a different relation skill.

CORE group	Acc.	Cov.	N	Reading
VG-compatible boxed retrieval	**0.591**	0.603	2045	Moderate transfer.
Attribute Binding, text-image	0.518	1.000	280	Best hybrid text cue.
Gaze-Attention	0.597	0.702	442	Stronger gaze transfer.
Occlusion-Depth	0.490	0.844	437	Depth direction is hard.
Spatial-Containment	0.612	0.724	343	Strongest reliable group.
Extreme Compositional OOD	0.710	0.246	31	High score, limited coverage.
Action-Role, text triplet	0.538	1.000	478	Weak role sensitivity.

Source: Project CORE retrieval logs; compositional evaluation reference: SugarCrepe.

Qualitative Result: "man riding horse"

This qualitative case shows the intended retrieval-facing output: predicted relation triplets can be inspected directly for a text query such as “man riding horse”.

Source: Project qualitative prediction example; retrieval motivation follows SGTR.

Conclusion

Conclusion

Model implementation

1. Implemented PURE as a compact directed-pair predicate scorer with dense CLIP evidence and object routing.
2. Added asymmetric SPOA relation states, progressive relation refinement, text/classifier scoring, relationness, and calibration readouts.
3. Trained and evaluated under a reproducible L4 24GB configuration.

Source: Summary based on project implementation and experimental logs; protocol context follows Neural Motifs.

Empirical outcomes

1. Reported VG150 PredCls with both R@50 and mR@50, separating raw and calibrated scores.
2. Built CORE examples and retrieval diagnostics for role, attribute, spatial, gaze, occlusion, and OOD cases.
3. Demonstrated relation-aware retrieval with predicted triplets such as (man, ride, horse).

Source: Summary based on VG150 logs, CORE diagnostics, and qualitative predictions; compositional context follows SugarCrepe.

Limitations and Future Work

Limitations

1. Grounding–bias trade-off: stronger grounding and calibration can improve tail-class fairness but may reduce aggregate R@50.
2. Static-image scope: the current system does not model temporal dynamics for video-based actions.

Source: Summary based on project implementation and experimental logs; long-tail context follows TDE.

Future works

1. Improve long-tail mR@50 with stronger debiasing and sample-level training.
2. Replace fixed calibration sweeps with constrained learnable calibration.
3. Extend object-label prediction for SGCls and SGDet settings.
4. Expand CORE vocabulary, audits, and retrieval demo coverage.
5. Explore 3D and video SGG after the 2D relation protocol is stable.

Source: Future directions informed by CFA, SBG, SGTR, and the report conclusion.

Retrieval Demo

A user enters a relation query such as “man ride horse”. The system matches the query against predicted triplets and returns images ranked by relation agreement.

Source: Retrieval application follows the SGG-to-structured-query motivation in SGTR.

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Thank you!